

Labsheet-1

Real-World Data Collection, Cleaning and Preprocessing using Python

```
In [3]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder, StandardScaler

# 1. Load dataset
df = pd.read_csv("WA_Fn-UseC_-HR-Employee-Attrition-selected-columns.csv")

# 2. Inspect
print(df.head())
print("\nShape:", df.shape)
print("\nMissing Values:\n", df.isnull().sum())

# 3. Handle missing values
for col in df.select_dtypes(include=np.number):
    df[col] = df[col].fillna(df[col].median())

for col in df.select_dtypes(include="object"):
    df[col] = df[col].fillna(df[col].mode()[0])

# 4. Remove duplicates
df.drop_duplicates(inplace=True)

# 5. Clean categorical values
for col in df.select_dtypes(include="object"):
    df[col] = df[col].str.strip()

# 6. Feature Engineering
if "Age" in df.columns:
    df["AgeGroup"] = pd.cut(
        df["Age"],
        bins=[0, 25, 35, 45, 60, 100],
        labels=["Young", "Early", "Middle", "Senior", "Very Senior"]
    )

# 7. Outlier detection and treatment
for col in ["Age", "MonthlyIncome", "TotalWorkingYears"]:
    if col in df.columns:
        Q1 = df[col].quantile(0.25)
        Q3 = df[col].quantile(0.75)
        IQR = Q3 - Q1
        df[col] = df[col].clip(
            Q1 - 1.5 * IQR,
            Q3 + 1.5 * IQR
        )

# 8. Label Encoding
le = LabelEncoder()

for col in ["Attrition", "Gender", "OverTime"]:
    if col in df.columns:
        df[col] = le.fit_transform(df[col].astype(str))
```

```
# One-Hot Encoding
df = pd.get_dummies(df, drop_first=True)

# 9. Standardization
cols = [
    col for col in ["Age", "MonthlyIncome", "TotalWorkingYears"]
    if col in df.columns
]

if cols:
    df[cols] = StandardScaler().fit_transform(df[cols])

# 10. Save dataset
df.to_csv("IBM_HR_Preprocessed.csv", index=False)

# Final output
print("\nPreprocessing Completed!")
print("Final Shape:", df.shape)
print("Missing Values:", df.isnull().sum().sum())
print("\nFinal Data:")
print(df.head())
```

	Age	Attrition	BusinessTravel	DailyRate	Department \
0	41	Yes	Travel_Rarely	1102	Sales
1	49	No	Travel_Frequently	279	Research & Development
2	37	Yes	Travel_Rarely	1373	Research & Development
3	33	No	Travel_Frequently	1392	Research & Development
4	27	No	Travel_Rarely	591	Research & Development

	DistanceFromHome	Education	EducationField	EmployeeCount	EmployeeNumber
0	1	2	Life Sciences	1	1
1	8	1	Life Sciences	1	2
2	2	2	Other	1	4
3	3	4	Life Sciences	1	5
4	2	1	Medical	1	7

Shape: (1470, 10)

Missing Values:

Age	0
Attrition	0
BusinessTravel	0
DailyRate	0
Department	0
DistanceFromHome	0
Education	0
EducationField	0
EmployeeCount	0
EmployeeNumber	0

dtype: int64

Preprocessing Completed!

Final Shape: (1470, 20)

Missing Values: 0

Final Data:

	Age	Attrition	DailyRate	DistanceFromHome	Education	EmployeeCount \
0	0.446350	1	1102	1	2	1
1	1.322365	0	279	8	1	1
2	0.008343	1	1373	2	2	1
3	-0.429664	0	1392	3	4	1
4	-1.086676	0	591	2	1	1

	EmployeeNumber	BusinessTravel_Travel_Frequently \
0	1	False
1	2	True
2	4	False
3	5	True
4	7	False

	BusinessTravel_Travel_Rarely	Department_Research & Development \
0	True	False
1	False	True
2	True	True
3	False	True
4	True	True

	Department_Sales	EducationField_Life Sciences	EducationField_Marketing \
0	True	True	False
1	False	True	False
2	False	False	False
3	False	True	False

4 False False False

	EducationField_Medical	EducationField_Other	\
0	False	False	
1	False	False	
2	False	True	
3	False	False	
4	True	False	

	EducationField_Technical	Degree	AgeGroup_Early	AgeGroup_Middle	\
0		False	False	True	
1		False	False	False	
2		False	False	True	
3		False	True	False	
4		False	True	False	

	AgeGroup_Senior	AgeGroup_Very Senior
0	False	False
1	True	False
2	False	False
3	False	False
4	False	False

Questions

1. Why is data preprocessing important?

Data preprocessing is an important step in data analytics because it converts raw data into a clean and useful format. It helps handle missing values, duplicates, errors, and inconsistent data, which improves the quality and reliability of analysis and machine learning models.

2. Explain different methods of handling missing values with examples.

Missing values can be handled using different methods:

- **Mean:** Replace missing numerical values with the average value.
- **Median:** Replace missing values with the middle value and is useful when outliers are present.
- **Mode:** Replace missing categorical values with the most frequently occurring value.
- **Deletion:** Remove rows or columns when the number of missing values is small or very large.

3. Differentiate between Label Encoding and One-Hot Encoding.

Label Encoding	One-Hot Encoding
Converts categories into numerical values.	Creates separate columns for each category.
Uses a single column.	Uses multiple columns.
Suitable for ordinal or binary categories.	Suitable for nominal categories.
Example: Male = 0, Female = 1.	Example: Male = [1,0], Female = [0,1].

4. What are outliers? How can they affect analytical results?

Outliers are data values that are significantly different from the majority of observations. They can affect the mean, standard deviation, correlation, and other statistical calculations. Outliers can also reduce the performance of some machine learning models.

5. Explain the difference between normalization and standardization.

- **Normalization:** Converts data into a fixed range, usually between 0 and 1.
- **Standardization:** Transforms data so that it has a mean of 0 and a standard deviation of 1.

Normalization is useful when a fixed range is required, while standardization is commonly used for machine learning algorithms that are sensitive to feature scale.

6. Why should duplicate records be removed before analysis?

Duplicate records can give extra importance to repeated observations and may produce incorrect statistical results. Removing duplicates ensures that each observation is counted properly and improves data consistency.

7. What is feature engineering? Give two practical examples.

Feature engineering is the process of creating new meaningful features from existing data to improve analysis or machine learning performance.

Examples:

- Creating **Age Group** from the `Age` column.
- Creating **Income Category** from the `MonthlyIncome` column.

8. Which preprocessing techniques would you apply to the IBM HR Employee Attrition dataset and why?

The following techniques can be applied:

- **Missing value treatment** to handle incomplete records.
- **Duplicate removal** to maintain data consistency.
- **Categorical encoding** to convert text values into numerical values.
- **Outlier detection and treatment** to reduce the effect of extreme values.
- **Standardization** to bring numerical features to a common scale.
- **Feature engineering** to create useful attributes for further analysis.

9. How does poor-quality data affect machine learning model performance?

Poor-quality data can reduce model accuracy and reliability. Missing values, duplicates, incorrect values, and outliers can introduce noise and bias. As a result, the model may produce inaccurate predictions and perform poorly on new data.

10. Name any three Python libraries commonly used for data preprocessing.

1. **Pandas** – Used for data loading, cleaning, and manipulation.
2. **NumPy** – Used for numerical calculations and array operations.

3. **Scikit-learn** – Used for encoding, scaling, normalization, and machine learning preprocessing.